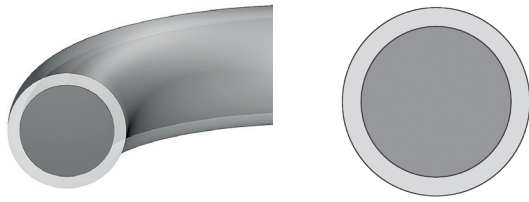


OR-11



Description

Encapsulated O-rings are cost-effective, high-performance seals for many critical applications. They consist of an elastomer core and a seamless FEP jacket. The elastomer core made of FKM or Silicone provides elasticity and pretension. The FEP jacket provides high resistance to aggressive media and ensures reliable sealing action.

Properties

FEP encapsulated O-rings combine the elasticity and sealing force of an elastomer with the excellent chemical resistance of PTFE.

- more elastic than PTFE + less expensive than FFKM
- better compression set and lower cold flow compared with PTFE
- higher chemical resistance than standard elastomers
- resistant to pressure and deformation
- wide thermal application range
- suitable for use in foodstuffs
- low friction, non-adhesive surface
- low gas permeability

Applications:

FEP encapsulated O-rings are used in all industrial sectors:

food processing industry, chemical process industry, gasoline transportation, petrochemical industry, paint + dye manufacturing, polymer production, etc.

- Covers
- Pumps, valves
- Couplings
- Pipelines
- Compressors
- Heat exchangers
- Vessels for chemicals
- Filters
- Mechanical seals

Materials

- FEP / VMQ
- FEP / FKM

FEP transparent, jacket

- FEP is the meltable version of PTFE
- FEP has the same chemical resistance and non-stick properties as PTFE
- chemically inert, resistant to almost all chemicals and solvents
- excellent thermal stability
- resistant to environmental influences such as ozone, sunlight and UV light
- compliant with FDA regulation

VMQ 70 red, core material

- excellent low temp. flexibility
- wide thermal application range
- excellent physiological properties
- excellent ageing resistance
- good resistance to animal and vegetable oil and fat,
- aliphatic lubricants, water up to 100 °C (212 °F)
- compliant with FDA regulation

FKM 70 black, core material

- very good resistance to chemicals
- very good thermal resistance
- very low gas permeability
- low compression set
- excellent ageing resistance
- good resistance to mineral oil based lubricants, fuels, aliphatic, aromatic and chlorinated hydrocarbons, HFD fluids, many aggressive chemicals and organic solvents

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Temperature ranges

FEP / VMQ: -60 °C to +200 °C -76 °F to +392 °F

FEP / FKM: -20 °C to +200 °C -4 °F to +392 °F

Temperature ranges are limit values and must always be considered in combination with the medium to be sealed and the operating pressure.

Approvals / Conformities

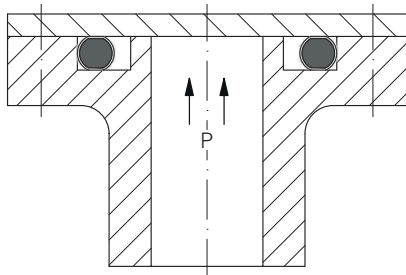
material	FDA	EC 1935/2004	REACH	RoHS	ADI free
FEP	X	X	X	X	X
VMQ	X		X	X	X
FKM			X	X	X

Assembly

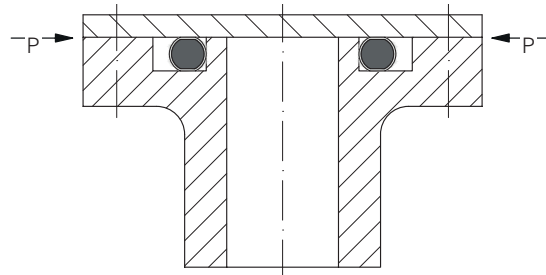
Please follow these guidelines for damage-free installation!

- right installation → trouble-free function
- damage-free
- before the assembly of the seal all components have to be cleaned
- housing surfaces should be lubricated
- sharp edges have to be deburred and rounded
- sharp edges like threads, feather key grooves, bore holes have to be covered
- the machine parts have to be chamfered
- bending and overstretching has to be avoided
- if O-rings have to be expanded over a bigger diameter, the elongation limits have to be regarded
- for more elasticity during the installation, O-rings should be heated up to 80 °C (176 °F) in hot water
- installation immediately after heating

Easy installation in axially open grooves,
e.g. flanges



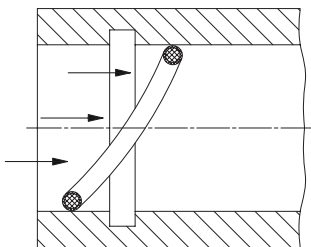
Pressure from inside



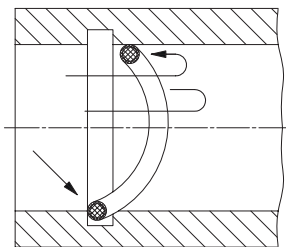
Pressure from outside

Installation in closed grooves:
radial sealing, inside sealing

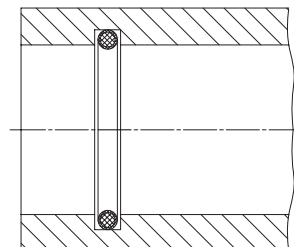
Step 1



Step 2



Step 3



radial sealing, outside sealing

